Boltzmann Transport from Observer-Patch Consistency: a Theorem-Conditional Reduction

OPH cosmology program

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Abstract

This note derives the general-relativistic photon–baryon Boltzmann transport used in CMB computations *inside* the OPH branch, as a theorem-conditional reduction: given the Einstein-branch geometry theorem, the recovered U(1) electromagnetic sector, the independently assumed Maxwell action and positive-residue physical photon pole, and the reversible repair-register structure, the observer-facing photon record dynamics obeys the standard Liouville equation with the Thomson collision functional, and its linearization on the capacity-closure FRW background is exactly the standard multipole hierarchy. OPH does not modify radiative transfer at first order; it *inherits* it. The value of the derivation is provenance: it identifies precisely (i) which OPH theorems supply each ingredient, (ii) the two places where OPH-specific kernels could enter (a number-changing repair-exchange channel Γ_{rec} and a collar-packet stress B_A), and (iii) the irreducible imports (absolute electron mass, recombination history) that any physical-CMB receipt must declare. One new finite-lattice receipt (BOLTZ-R1, refinement Liouville) is defined and is testable in the public simulator. This note does not close the physical CMB prediction gate; it reduces that gate to a finite, named list.

1 Claim boundary

Everything below is conditional on the branch theorems it cites. The derivation upgrades the status of the *transport form* used by the diagnostic CAMB lane from "imported wholesale" to "theorem-conditional reduction with declared imports." It does not certify the initial spectrum (amplitude or tilt), the recombination history, the absolute Thomson normalization, or any likelihood statement. Those remain the gates tracked in issues #330, #522, #374, and #363 respectively. No statement in this note reads Planck data.

2 Branch inputs

Assumption 2.1 (A-GEOM: Einstein branch)**.** *The observer-facing normal form carries a Lorentzian metric g_{ab} satisfying $G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{geom}}\langle T_{ab}\rangle$, with $G_{\mathrm{geom}} = \ell_\star^2$ from the edge-entropy/area-law theorem and $\Lambda\ell_\star^2 = 3\pi/N_\star$ from capacity closure. $N_\star$ is the SL-4 estimate read from measured Λ with the scale bridge of the strange-loop principle register; Λ is an input to this assumption, not an OPH output. Promotion of $N_\star$ beyond the SL-4 estimate runs through generator G2 of the project consistency stack: constructing the readback map F and its contraction certificate. Provenance: branch theorem (paper stack A22/A5); not re-proved here.*

Assumption 2.2 (A-QED: electromagnetic action and photon-pole receipt)**.** *The realized branch carries the Standard Model quotient and the Maxwell kinetic action with a* U(1) *vertex of strength*

$\alpha = \alpha_{\rm root}$ *(source-side). In addition, this transport note assumes a positive-energy physical QED quantization with two transverse states and a positive-residue isolated pole at* $k^2 = 0$ *on the ordinary Lorentz-vacuum, unbroken electromagnetic phase. This extra photon-pole receipt is not implied by the reconstructed abstract group. Charged-lepton* ratios *are branch outputs; the absolute electron mass* m_e *is empirically anchored. Provenance: branch theorem (A20/A21), the explicit quantum-pole assumption just stated, and one declared import (*m_e*).*

Assumption 2.3 (A-BG: background)**.** *The homogeneous limit of the branch is spatially flat FRW,* $\mathrm{d}s^2 = a^2(\tau)\,[-\mathrm{d}\tau^2 + \delta_{ij}\mathrm{d}x^i\mathrm{d}x^j]$*, with Friedmann dynamics from Assumption 2.1. Scalar perturbations are written in the conformal Newtonian gauge,*

$$\mathrm{d}s^2 = a^2(\tau)\left[-(1+2\psi)\,\mathrm{d}\tau^2 + (1-2\phi)\,\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j\right],$$

with conventions of Ma–Bertschinger (1995). Overdots are ∂_τ*.*

Assumption 2.4 (A-REC: recombination import)**.** *The free-electron history* $n_e(\tau)$ *(equivalently* $x_e(\tau)$*) is computed from standard atomic physics (Saha/Peebles-class) using imported atomic data. This is the same import class as the cesium-clock wall in the gravity stack: it contains no CMB-target information, so it is a permissible declared input under the anti-fitting rule, but it is an import, not a branch output.*

Assumption 2.5 (A-INIT: initial conditions (gated))**.** *Initial curvature perturbations* $\zeta(\mathbf{k})$ *are Gaussian with spectrum* $\Delta^2_\zeta(k) = A_s\,(k/k_*)^{n_s-1}$*. The Gaussian and statistically isotropic character is the MaxEnt screen-covariance theorem; the tilt certificate is issue #522; the amplitude and the radial screen-to-k lift are issue #330. This note treats* A_s*,* n_s *as gated inputs and derives nothing about them.*

3 Theorem B1: reversible repair gives Liouville transport

Definition 3.1 (One-photon record density)**.** *Conditional on the physical pole in Assumption 2.2, let* $\mathcal{E}_\gamma$ *be its positive-energy photon sector. Between record commits, a single excitation is a record continued across patch overlaps. Coarse-graining patch position and overlap momentum labels defines the phase-space record density* $f(x^i, q, \hat{n}, \tau)$ *on the forward mass shell, where* q *is comoving momentum magnitude and* $\hat{n}$ *the direction.*

Theorem 3.2 (Refinement Liouville)**.** *On a branch passing the simulation test, with (i) the Maxwell action, quantum photon-pole receipt, and overlap Lorentz kinematics (A-QED, A-GEOM), and (ii) repair moves organized in reversible write/check pairs (the orientation doubling in* $m_{\rm rep} = 2\dim(\mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)) = 24$*), the between-commit record flow satisfies, in the refinement limit,*

$$\frac{Df}{\mathrm{d}\tau} \;\equiv\; \frac{\partial f}{\partial\tau} + \frac{\mathrm{d}x^i}{\mathrm{d}\tau}\frac{\partial f}{\partial x^i} + \frac{\mathrm{d}q}{\mathrm{d}\tau}\frac{\partial f}{\partial q} + \frac{\mathrm{d}\hat{n}}{\mathrm{d}\tau}\cdot\frac{\partial f}{\partial\hat{n}} \;=\; C[f],$$

where the characteristics are the null geodesics of g_{ab} *and* $C[f]$ *is supported on record commits only. In particular the collisionless flow preserves the Liouville measure.*

Proof sketch, with the open receipt named. Characteristics. A single-excitation record continued across an overlap must agree with both patches' readouts; by overlap Lorentz kinematics the unique continuation consistent with all overlapping patches is parallel transport along the branch metric's null geodesic. Any other continuation produces a nonzero overlap mismatch, which the repair fixed point sets to zero; schedule independence (confluence) makes the continuation unique and schedule-free. Hence $\mathrm{d}x/\mathrm{d}\tau$, $\mathrm{d}q/\mathrm{d}\tau$, $\mathrm{d}\hat{n}/\mathrm{d}\tau$ are the geodesic-flow coefficients; on the background of Assumption 2.3, $\mathrm{d}q/\mathrm{d}\tau = q[\dot{\phi} - \ldots]$ takes the standard form used in §5.

Measure preservation. Each repair move is a write/check pair: a bijection on the finite record configuration space with an explicit inverse (this is what the orientation doubling tracks). A finite composition of bijections has unit Jacobian on the induced finite phase space, for every schedule; confluence makes the endpoint schedule-independent, so the coarse-grained flow inherits unit Jacobian. The refinement limit of a sequence of unit-Jacobian flows converging to the geodesic flow preserves the Liouville measure.

Open receipt. The convergence step, that the finite patch flow's coarse-graining converges to the geodesic flow with vanishing dispersion, is exactly the kind of statement the abstract Lean layer proves in its setting (approximate schedule independence, refinement naturality) but has not been instantiated for the photon sector. We name it BOLTZ-R1 and give a simulator test in §7. Until BOLTZ-R1 is discharged, Theorem 3.2 is theorem-conditional. □

Remark 3.3. The physical content is that *irreversibility enters only at commits.* Free streaming is unitary bookkeeping; entropy production is localized in the record-commit events that the collision functional counts. This is the transport-level face of the general OPH position that measurement-like irreversibility lives at the record interface.

4 Theorem B2: record commits give the Thomson collision functional

Theorem 4.1 (Scalar-intensity commit-collision functional)**.** *Let commits between photon-class and electron-class records occur with the soft-limit matrix element of the recovered* U(1) *vertex (A-QED). Then, to first order in perturbations and at low occupancy, the collision functional in Theorem 3.2 is the Thomson functional: in the electron rest frame,*

$$C[f](q,\hat{n}) \;=\; a\, n_e\sigma_T \left[\frac{3}{16\pi}\int \mathrm{d}\Omega'\,(1+(\hat{n}\cdot\hat{n}')^2)\,f(q,\hat{n}') - f(q,\hat{n})\right], \qquad \sigma_T = \frac{8\pi}{3}\frac{\alpha^2}{m_e^2},$$

boosted to the frame where baryons move with velocity v_b. Moreover the write/check reversibility of commits implies detailed balance of the angular commit kernel. For every fixed photon energy q, isotropic angular distributions are stationary under this Thomson operator. This displayed scalar-intensity kernel does not by itself derive the Stokes-space Thomson matrix needed for polarization.

Proof sketch. At low occupancy the commit rate between momentum cells is bilinear in the occupancies with kernel given by the squared vertex; the soft (photon energy $\ll m_e$) limit of the recovered U(1) vertex is classical Thomson scattering with the stated total cross-section and $1+\cos^2$ angular kernel; this step is standard QED evaluated on the branch couplings and is not repeated here. Detailed balance: a write/check pair and its orientation reverse implement the commit and its inverse with equal kernel weight; a symmetric kernel damps angular anisotropy at each fixed q. It does not mix frequencies, so its stationary kernel contains every isotropic spectrum $f(q)$ rather than a unique Bose or Planck occupancy. Momentum conservation per commit, summed over commits, gives equal and opposite momentum transfer densities to the electron–baryon fluid, with coefficient fixed by $\bar{\rho}_\gamma/\bar{\rho}_b$ bookkeeping; this is the Compton drag used in §5. □

Remark 4.2 (Polarization provenance). The polarization hierarchy used below imports the standard polarized Thomson scattering matrix acting on Stokes brightness. The scalar $1+\cos^2\theta$ kernel displayed in Theorem 4.1 is its intensity projection, not a derivation of the Q/U collision terms. A source-level OPH derivation must construct the polarized commit kernel from the recovered vertex and discharge that extra bridge; until then the polarization sector and the associated 16/15 diffusion factor are textbook imports rather than consequences of the displayed operator.

Remark 4.3 (Blackbody provenance). The Planck background spectrum is an initial-condition and thermal-history import in this note, not a consequence of the displayed Thomson functional. Even recoil-inclusive Comptonization conserves photon number and generically approaches a Bose–Einstein spectrum with a chemical potential. Selecting the zero-chemical-potential Planck spectrum requires the appropriate early thermalization history, including photon-number-changing channels such as double Compton scattering and bremsstrahlung, or an independently justified blackbody initial condition. An OPH derivation of the blackbody spectrum requires those channels and their rates rather than inferring spectral uniqueness from angular detailed balance.

Remark 4.4 (Provenance of the normalization). $\alpha = \alpha_{\rm root}$ is source-side. m_e is an import (Assumption 2.2): OPH supplies no source-derived charged-lepton absolute mass, so σ_T's normalization is import-tagged (Boltz-N1 in the ledger). $n_e(\tau)$ is import-tagged by Assumption 2.4 (Boltz-X1).

Remark 4.5 (OPH-specific channels). Two and only two places exist where OPH kernels can modify this functional at linear order: (i) a number-changing repair-exchange channel $\Gamma_{\rm rec}(k,a)$ adds a monopole source and a spectral-distortion term; it enters $C[f]$ additively and is gated by the #374 receipts (active fiber, physical clock, common-parent response); (ii) a collar-packet stress $\rho_A, B_A(k,a)$ enters only the gravitational source side (the Einstein constraints in §5), never $C[f]$. With both gates closed at zero, the transport below is exactly standard.

5 Theorem B3: the linear multipole hierarchy

Theorem 5.1 (Reduction to the standard hierarchy)**.** *Linearize Theorems 3.2–4.1 on the background of Assumption 2.3 with scalar perturbations, expand the temperature brightness $F_\gamma(\mathbf{k},\hat n,\tau)$ and polarization brightness G_γ in Legendre multipoles with $\mu = \hat k\cdot\hat n$. Writing $\delta_\gamma = F_{\gamma 0}$, $\theta_\gamma = \frac{3}{4}kF_{\gamma 1}$, $\sigma_\gamma = \frac{1}{2}F_{\gamma 2}$, and $\dot\kappa \equiv a n_e \sigma_T$, the system is (Ma–Bertschinger 1995, eq. 63):*

$$
\begin{aligned}
\dot\delta_\gamma &= -\tfrac{4}{3}\theta_\gamma + 4\dot\phi,\\
\dot\theta_\gamma &= k^2\left(\tfrac{1}{4}\delta_\gamma - \sigma_\gamma\right) + k^2\psi + \dot\kappa\,(\theta_b - \theta_\gamma),\\
2\dot\sigma_\gamma &= \tfrac{8}{15}\theta_\gamma - \tfrac{3}{5}kF_{\gamma 3} - \tfrac{9}{5}\dot\kappa\,\sigma_\gamma + \tfrac{1}{10}\dot\kappa\,(G_{\gamma 0} + G_{\gamma 2}),\\
\dot F_{\gamma\ell} &= \frac{k}{2\ell+1}\left[\ell F_{\gamma(\ell-1)} - (\ell+1)F_{\gamma(\ell+1)}\right] - \dot\kappa F_{\gamma\ell}, \qquad \ell \ge 3,\\
\dot G_{\gamma\ell} &= \frac{k}{2\ell+1}\left[\ell G_{\gamma(\ell-1)} - (\ell+1)G_{\gamma(\ell+1)}\right] + \dot\kappa\left[-G_{\gamma\ell} + \tfrac{1}{2}\left(F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2}\right)\left(\delta_{\ell 0} + \tfrac{\delta_{\ell 2}}{5}\right)\right],
\end{aligned}
$$

coupled to baryons,

$$
\dot\delta_b = -\theta_b + 3\dot\phi, \qquad \dot\theta_b = -\frac{\dot a}{a}\theta_b + c_b^2 k^2\delta_b + k^2\psi + \frac{4\bar\rho_\gamma}{3\bar\rho_b}\,\dot\kappa\,(\theta_\gamma - \theta_b),
$$

and to the Einstein constraints of Assumption 2.1,

$$
k^2\phi + 3\frac{\dot a}{a}\left(\dot\phi + \frac{\dot a}{a}\psi\right) = -4\pi G a^2\,\delta\rho_{\rm tot}, \qquad k^2(\phi - \psi) = 12\pi G a^2(\bar\rho + \bar p)\sigma_{\rm tot},
$$

where $\delta\rho_{\rm tot}$ and $\sigma_{\rm tot}$ include the gated ρ_A, B_A contributions of Remark 4.5 (zero until their receipts pass). With $\Gamma_{\rm rec} = B_A = 0$ this is verbatim the standard Einstein–Boltzmann system.

Proof. For scalar intensity, insert the geodesic characteristics of Theorem 3.2 on the perturbed metric, the collision functional of Theorem 4.1, keep first order, and project on Legendre multipoles. The $G_{\gamma\ell}$ equations additionally use the imported polarized Thomson matrix stated above. The remaining steps are the textbook computation (Ma–Bertschinger 1995; Dodelson, *Modern Cosmology*); no OPH-specific term survives except the two gated channels, which we have displayed at their unique insertion points. □

6 Acoustic normal form, damping, and the line of sight

Proposition 6.1 (Acoustic oscillations). *In the tight-coupling regime $\dot\kappa \gg k,\ \frac{\dot a}{a}$, define $R \equiv 3\bar\rho_b/4\bar\rho_\gamma$ and $\Theta_0 = \delta_\gamma/4$. To leading order in $1/\dot\kappa$,*

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\Big[(1+R)\,\dot\Theta_0\Big] + \frac{k^2}{3}\,\Theta_0 = -\frac{k^2}{3}(1+R)\,\psi + \frac{\mathrm{d}}{\mathrm{d}\tau}\Big[(1+R)\,\dot\phi\Big],$$

a driven oscillator with sound speed $c_s^2 = 1/[3(1+R)]$. The WKB solution gives, at last scattering τ_*,

$$[\Theta_0+\psi](\tau_*)\ \propto\ \cos(k\,r_s(\tau_*)), \qquad r_s(\tau_*) = \int_0^{\tau_*} c_s\,\mathrm{d}\tau,$$

so extrema sit at $k_n\,r_s(\tau_*) = n\pi$: compressions (odd n) are baryon-enhanced, rarefactions (even n) suppressed, and the dark-matter driving of the potentials boosts the third peak: every feature of the measured TT curve is produced by this system and by no OPH-specific term.

Proposition 6.2 (Silk damping). *At next order in $1/\dot\kappa$ the oscillator acquires the diffusion envelope e^{-k^2/k_D^2} with*

$$k_D^{-2} = \int_0^{\tau} \frac{\mathrm{d}\tau'}{6(1+R)\,\dot\kappa}\left[\frac{R^2}{1+R} + \frac{16}{15}\right],$$

the $16/15$ arising from the polarization terms of Theorem 5.1; photon diffusion between commits is the physical mechanism, and its OPH reading is transparent: on scales crossed by a random walk of commit-free transport legs, the record field cannot maintain contrast.

Proposition 6.3 (Line-of-sight solution). *With visibility $g(\tau) = \dot\kappa\,e^{-\kappa(\tau,\tau_0)}$, the temperature multipoles today are*

$$\Theta_\ell(k,\tau_0) = \int_0^{\tau_0}\mathrm{d}\tau\,\Big\{g\left[\Theta_0+\psi+\tfrac{\Pi}{4}\right]j_\ell(k(\tau_0-\tau)) + \Big(\frac{g\,v_b}{k}\Big)^{\dot{}}\,j_\ell + \frac{3}{4k^2}(g\Pi)^{\ddot{}}\,j_\ell + e^{-\kappa}(\dot\psi+\dot\phi)\,j_\ell\Big\},$$

($\Pi = F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2}$; every unlabelled j_ℓ has argument $k(\tau_0-\tau)$), and

$$C_\ell^{TT} = 4\pi\int\frac{\mathrm{d}k}{k}\,\Delta_\zeta^2(k)\left|\frac{\Theta_\ell(k,\tau_0)}{\zeta(k)}\right|^2.$$

Given Assumption 2.5's gated (A_s, n_s) and the imports of Assumptions 2.2–2.4, this reproduces the curve computed by the diagnostic CAMB lane; agreement of an independent implementation with CAMB at the sub-percent level is receipt Boltz-V1.

7 Receipts and simulator tests

Receipt	**Statement**	**Status**
Boltz-R1	Finite-lattice refinement Liouville: coarse-grained photon-class record transport between commits converges to geodesic flow, schedule-independently, with vanishing measure defect under refinement.	open; simulator test below
Boltz-N1	σ_T normalization: absolute m_e is a declared import until a charged-family absolute-scale theorem exists.	declared import
Boltz-X1	$x_e(\tau)$ recombination history from imported atomic data; contains no CMB-target information.	declared import
Boltz-V1	Independent hierarchy implementation matches CAMB TT/TE/EE at stated tolerances with identical inputs.	open; engineering
Tilt certificate	Theorem-grade n_s (clock normalization).	issue #522
Amplitude + lift	A_s and radial screen-to-k map.	issue #330
Kernels	Γ_{rec}, ρ_A, B_A physical receipts.	issue #374
Likelihood	Official likelihood, provenance, and data-use contract.	issue #363

Simulator test for Boltz-R1. Inject a labeled low-weight record packet on the 64k lattice; propagate under N random repair schedules; measure the arrival-time and dispersion statistics of the packet across the screen. Pass condition: schedule-to-schedule divergence of the packet trajectory consistent with zero at the run's mismatch tolerance, and dispersion shrinking under refinement (16k→64k→256k) at the rate the refinement-naturality bounds predict. Fail condition: schedule-dependent transport, which would falsify Theorem 3.2 on the implemented branch.

8 What this note does and does not close

It records, conditionally on the cited branch theorems, the *scalar angular form* of the transport and the locations where the two gated OPH kernels would enter. The polarization hierarchy, blackbody background, absolute normalizations, and recombination history remain imported. Thus the physical-CMB prediction gate contains a source-level polarized collision bridge, blackbody/thermalization provenance, Boltz-R1, and Boltz-V1, in addition to the pre-existing gates (#330, #522, #374, #363). It does not certify initial conditions, normalizations, recombination, or likelihoods, and nothing here reads Planck data. Under the anti-fitting rule, the acoustic peak structure remains attributable to standard plasma physics; OPH's falsifiable exposure stays concentrated in the primordial inputs, exactly where issues #330 and #522 put it.

References

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- S. Dodelson and F. Schmidt, *Modern Cosmology*, 2nd ed., Academic Press (2020).
- U. Seljak and M. Zaldarriaga, *A Line-of-Sight Integration Approach to Cosmic Microwave Background Anisotropies*, ApJ 469, 437 (1996).
- W. Hu and M. White, *CMB Anisotropies: Total Angular Momentum Method*, Phys. Rev. D 56, 596 (1997).
- OPH paper stack: *Observers Are All You Need*; *Reality as a Consensus Protocol*; *Recovering Relativity and the Standard Model from Observer Overlap Consistency*; *OPH Cosmology Finite-Source CMB Program* (this repository).